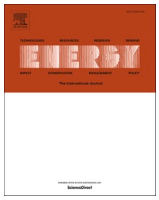
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ROSMOSE: A web-based decision support tool for the design and optimization of industrial and urban energy systems

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ABSTRACT

Energy efficiency is crucial for the sustainable operation of industrial and urban sectors. However, practicing engineers seldom have access to open-source tools that can readily evaluate and compare scenarios. In this work, a novel web-based tool called ROSMOSE is developed and proposed for analyzing the energy efficiency of industrial and urban systems and comparing potential process integration options. This optimization framework computes and graphically represents the minimum energy requirements, the pinch temperatures, and the Grand and integrated Carnot composite curves of process systems. Results are used to design utility systems that meet energy demands, while minimizing a specified objective function, such as total cost or environmental impacts. The Quarto environment, in which ROSMOSE is built upon, allows the integration of various open-source software and tools, such as database handling software, process modeling suites, and optimization solvers, along with interactive data visualization and reporting tools. This paper discusses the application of ROSMOSE for the energy integration and total site optimization of a dairy process consisting of milk treatment, and cream and cheese production to demonstrate the tool features. Other integration options, such as the use of heat pumps or solar panels, and the production of soft drinks or biogas as value-added byproducts, are also proposed and evaluated. As a result, from 75 % to 90 % in fuel savings and up to 80 % CO_2 emissions reduction in the dairy plant are identified by optimizing the utility systems integration. Moreover, complete electrification via heat pumps eliminates any fossil fuel needs and it results in 85 % energy savings and up to 96 % CO_2 emissions reduction. The activation of renewable electricity sources, such as solar power, made plausible this fully electrified scenario. Finally, optimized waste management strategies led to in-house fuel production and net export of biogas to the grid, but they also reduce cheese yield by prioritizing the biogas production from whey upgrading process. In this way, decision-makers have access to a clear performance comparison for a list of system configurations to draw informed trade-offs. A similar approach could be adopted to aid decision-making for the design and operation of other industrial and urban energy systems using ROSMOSE.

1. Introduction

Following decades of research and development, the increasing complexity of industrial processes and energy system integration problems led to the emergence of a series of heat integration computational tools and simulation environments. Despite the number of existing commercial software alternatives, their proprietary features, rigid user interfaces, and hidden algorithms offer reduced accessibility to a wider scientific community and render any data exchange between various computational platforms cumbersome. For this reason, the key outcome of this work is to make the ROSMOSE tool accessible to fellow researchers and industrial partners of the open-source community. In this section, the development history of energy integration methods is briefly introduced, and a comparison between selected tools for process and energy system integration, simulation, and optimization is presented. Next, the methods and functionalities, the handled formats, the

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superstructure-based simulation capacities, the mathematical formulation, and the data visualization routines in the ROSMOSE tool are detailed. The relevance of ROSMOSE is highlighted, and the advantages and limitations of the tool are reported.

A case study for energy integration and total site optimization of a dairy plant, encompassing milk treatment up to cream and cheese production, is presented. Several integration options are evaluated to valorize the waste streams of the dairy processes (e.g., whey and sludge), including the production of mesost (i.e., brown cheese), biogas, or soft drinks. The integration and optimization of different utility systems are studied to enhance the overall process efficiency and economic performance while reducing process environmental impact. Heat pumps are of particular interest in this application due to their compatibility with the energy demands of the food industry at moderate temperature levels (~120 °C) [1–3]. From an economic point of view, the dairy process case study is of regional interest, as in 2022, the European dairy market attained a value of €165 billion (i.e., 24 % of worldwide production), and global consumption rates reached 115 kg of dairy products per capita [4]. From an energy efficiency and environmental point of view, the food and beverage industry is the second most energy-consuming industrial sector in Switzerland, sharing 14 % of total final energy consumption in the Swiss industry and consequently emitting 14 % of domestic industrial CO_2 emissions [5]. Increasing the overall energy and economic efficiency of the dairy production sites and reducing their carbon footprint will directly result in a reduction of Swiss CO_2 emissions, which also stresses the relevance of this type of assessment by using openly distributed and transparent modeling and documentation tools like ROSMOSE. The supplementary material further describes the steps to access the open-source code.

1.1. Development history of energy integration methods

Higher energy prices and more competitive markets are pushing industrial and urban energy systems to adopt more efficient approaches, thus fostering resilience and sustainability in these sectors. Process integration methods aim to minimize energy consumption residues and operating expenditures by maximizing waste heat recovery and optimizing utility needs [6]. These methods, notably pinch analysis, have undergone significant evolution over the years to enhance energy efficiency in both industrial and urban systems. In the late 1970s, engineers exploring the design of heat exchanger networks (HENs), combining thermodynamic principles with heuristic rules, demonstrated the usefulness of the pinch point concept [7]. At the same time, in another independent study, Linnhoff and Flower [8] introduced the problem table method that allows calculating the minimum energy requirement of an industrial process and generating a graphical representation of the waste heat available at different temperature levels, hence devising the Pinch analysis method. Based on these ideas, Cerda et al. [9] analyzed ways to convey thermal energy between process streams to obtain the minimum utility costs and devise optimal HEN designs. Later, Papoulias and Grossmann published a methodology for a complete process system integration consisting of a chemical process plant, a heat transfer network, and a utility system [10]. The problem was solved simultaneously using a mixed integer linear programming (MILP) formulation. Further developments by Maréchal and Kalitventzeff proposed a novel graphical way for separately representing the composite curve of the utility and process streams in the so-called integrated composite curve (ICC) [11]. The strategy starts with the utility system integration; next, it calculates the heat load distribution (HLD) and synthesizes the topology of the HEN [12]. The work was extended to solve industrial-scale problems involving combined heat and power production (CHP) using a steam network superstructure [13]. State-of-the-art problem-solving for process optimization adopts the MILP approach due to the many advantages offered by the linearity of the problem and the effectiveness of available solvers [14]. One drawback of such a methodology is the impossibility of modeling non-linear effects without prior simplification. However, unless dynamic simulation and real-time optimization are intended, MILP formulations are acceptable to assess system integration problems for which binary and continuous optimization variables can be used to capture the steady-state performance. Hence, the objectives of this study are to.

- Develop a user-friendly tool (ROSMOSE) for the optimization and total site integration of industrial and urban energy systems, following pinch analysis and other energy integration methods, and make it available for researchers and industrial partners.
- Demonstrate the key features of ROSMOSE using a sample dairy process as the main case study.
- Promote the shareability of knowledge by using this newly developed tool and enable interactive future developments based on feedback and collaboration from a wider scientific community.

1.2. Review of existing computational tools

Some examples reported in the literature include Aspen Energy Analyzer®, gPROMS®, i-Heat®, and hiTRAN®. Aspen Energy Analyzer® is an Aspentech® energy management software used to perform optimal HEN design to minimize process energy consumption [15]. Advantages of this software include its compatibility with flowsheeting simulators, like Aspen HYSYS® and Plus®, and its easy-to-use interactive environment. Another suite fully incorporated within Aspentech® flowsheeting simulators is Aspen Exchanger Design & Rating®, which produces options for optimal HEN designs with specifications of different types of heat exchangers, such as shell-and-tube, plate, or air-cooled exchangers [16]. On the other hand, gPROMS® is an alternative flowsheeting software that offers flexibility in defining the governing mathematical equations for process units [17]. However, it is not focused on the total site optimization of multiple process units or any integration of industrial and urban energy systems. Another program called i-Heat® is a development from Process Integration Limited® for the design, retrofitting, and optimization of HENs, which enables the identification of operating conditions for reduced energy consumption [18]. A common limitation in using most of these software alternatives is their commercial license-based features with rigid user interfaces (UI) and hidden algorithms, which hinder data exchange with other platforms and prohibit modifications to the source code. In fact, open-source accessibility and communication among different software are major challenges in using existing process integration tools for designing and optimizing the overall energy consumption of industrial systems [19, 20].

Howells et al. developed a software called “OSeMOSYS” which is an open-source energy modeling system targeted toward policymakers and analysts. The software gives a global or country-specific view of energy system scenarios for variable inputs and operating conditions [21]. Studies on the electricity system model connections have recently been published using this tool, providing an illustrative workflow example [22]. However, the scope of this tool does not encompass smaller-scale industrial and urban systems optimization or heat and mass integration principles. It is focused on energy supply technologies rather than energy-efficient use and conversion. OSeMOSYS is maintained by a community of practitioners, universities, and governmental entities named OpTIMUS, which provides open-source tools for energy policy and driving climate transitions [23]. Other researchers have published similar work related to the optimization of energy systems from varying perspectives. One of which is the work of Jing et al. who conducted a parametric study to evaluate the most influencing conditions on energy efficiency and electricity generation [24]. However, this approach does not focus on pinch analysis or heat integration but rather leverages non-linear optimization methods (e.g., NSGA-II) and proprietary environments (e.g., MATLAB and EES), reducing the reproducibility and shareability of the results.

On the other hand, Zoder et al. proposed an open-source Python-

based framework to promote object-oriented programming for modeling energy conversion processes [25]. The work focused on gas and steam turbine cycles for power generation and was linked to the Coolprop database to estimate the fluid properties. However, the open-source tool does not address the total site integration problem for more complex industrial and chemical plants. Another approach to maximize heat recovery potential in industrial systems uses computation fluid dynamics (CFD) to study heat transfer phenomena, enabling waste heat recovery from various systems, such as mini-channel heat sinks [26]. This approach complements process system engineering but differs in the scope, targets, and type of results obtained. Other Software options, such as EnergyPlus developed by the U.S. Department of Energy, are used by researchers to perform optimization of photovoltaic and heat pump integrated systems in various climatic locations across Morocco [27].

Lastly, PinCH is a commercially available software for implementing pinch analysis in industrial applications derived from PinchLENI, originally developed at EPFL [28,29]. However, unlike the ROSMOSE tool proposed in this work, the interface to Excel for calculating the energy requirements of individual basic processes is still under development. PinCH software is a subscription-based tool focused on identifying pinch analysis targets and proposing HEN designs based on predefined utility systems. Other tools, such as Simulus Pinch, INTEGRATION, and Heatit, make part of the list of process modeling and integration frameworks, and their characteristics are compared in Table 1 with more prominent energy integration software available in the market and published literature.

Table 1
Summary of prominent modeling and simulation, and energy integration computational tools.

Tool	Pros	Cons	Ref.
Aspen Energy Analyzer®	• Compatibility with Aspen flowsheeting simulators • Interactive environment	o Reduced exchange capabilities with other software o Hidden algorithms and relatively fixed solver settings	[15]
i-Heat®	• Focused on HEN design and retrofit • Interactive environment	o Rigid reporting based on existing template o Proprietary software with hidden algorithms	[18]
OSeMOSYS	• Open-source energy modeling • Targeted for driving policy change	o Total site and equipment-wise optimization capabilities for urban clusters and industrial plants is not included	[22]
PinCH	• Analyze storage and discontinuous batch systems • Tailor-made functionalities only by contacting developers	o Limited technologies in developer-defined list o Proprietary software with hidden algorithms	[28]
ProSimPlus Energy	• Specialized exergy analysis tool • Compatible with other Simulus thermodynamic software	o Proprietary software with hidden algorithms o Only exergy audit, no system utility integration	[30]
INTEGRATION	• User-friendly interface and comprehensive guide • Well-established and documented output projects	o Exclusively offered to Canadian engineers attending a specific training course o Limited technologies list predefined by developers	[31]
Heatit by PINCHCO	• Published Excel-sheet for practicing engineers • Many industrial applications reported	o No utility integration or system optimization, but only pinch analysis results o Calculation and plotting tabs are locked (uneditable)	[32]

1.3. Advantages and limitations of ROSMOSE tool

One of the main drawbacks of existing process integration tools is related to their lack of open-source accessibility, which hinders the contribution and adaptation of the methods and algorithms used for performing comparative, multi-scale energy integration analyses of a set of industrial plants, decarbonization technologies, and urban and energy systems (i.e., grids). Moreover, the rigid reporting capabilities of proprietary and some open-source tools contrast with the versatility of reproducible and customizable scientific reports, making the incorporation of flexible and interactive reporting tools which are compatible with several process flowsheeting environments and mathematical optimization algorithms necessary.

To tackle these issues, a flexible and integrated open-source tool for the synthesis and design of complex energy systems, called OSMOSE, was devised [33]. The tool undergoes continuous development at the Industrial Process and Energy System Engineering (IPESE) group, capitalizing on more than 20 years of research to expand its capabilities, robustness, and versatility [34]. Lua programming language is used to write the source code of OSMOSE, as it features a fast and extensible architecture [35]. More recently, the OSMOSE engine was integrated into the Quarto environment to conceive a high-level language called ROSMOSE. This newly developed language is used in a novel open-access platform that combines modeling, simulation, data analysis, and rigorous process integration methods with a versatile visualization and documentation scheme to write reproducible dynamic reports while embedding code chunks and generating results, all in one platform [36]. The models of the industrial processes and energy technologies can be user-defined or imported from well-established flowsheeting software, such as Aspen Plus ® and DWSIM, among others. An example of this functionality is explained in more detail in the Methods section of this manuscript. Moreover, the thermophysical properties of other substances can be computed through a Python wrapper of the Coolprop library [37]. The modeling options of the integrated utility systems can be programmed either as energy conversion units with specific user-defined inputs or information from vendors' datasets or they can be adapted from user-friendly templates based on rigorous modeling superstructures of steam network and heat pumping technologies. The set of energy technologies defined by the previous models can be used to perform the energy integration procedures while solving the total site cost minimization problem. A rigorous design of those waste heat recovery systems is key for revealing hidden potentials that other process simulation tools may leave untapped, for instance, when typical utility systems (boiler, cooling water, etc.) and utility streams with generic temperature and pressure levels are integrated. A combination of cogeneration and heat pump systems is relevant to anticipate stringent scenarios of intensive industrial decarbonization via electrification for supplying low-temperature heat to both residential and industrial applications.

As for ROSMOSE data outputting capabilities, the formats of the modeling and optimization results can be either static (e.g., portable document format – PDF) or interactive (e.g., hypertext markup language – HTML, useful for web-based reports), which can be accessed through typical open-source web browsers and IDEs [38]. Reader-interactive options can be realized via application programming interfaces, such as Mermaid [39] and Shiny [40]. Moreover, the user models and output reports are equipped with a flexible bibliography data format (.bib) and management system that enables well-referenced programming and reporting. All the aforementioned functionalities can hardly be found at the same time in any of the existing open-source or proprietary software previously described. As a result, the versatility of the ROSMOSE tool confers advantages in terms of process synthesis, optimization, reporting, sharing, and documentation, which are in agreement with the target of performing inclusive, community-oriented, and transparent industrial

process and energy system assessment for the development of energy transition technologies and pathways. The source code is made available and the instructions for running an example code are provided in the Supporting Information SI-1 and SI-2. In this paper, a detailed explanation of the mechanisms of this newly developed tool is presented, along with an illustrative example that showcases how ROSMOSE can aid decision-making for the synthesis, design, and optimization of industrial and urban energy systems.

Besides its advantages, ROSMOSE also may have limitations in terms of maintainability, modeling accuracy, and programming issues. In effect, an open-source contributing community is fundamental to maintaining, curating, and promoting the utilization and development of the ROSMOSE tool. For this reason, a robust infrastructure is necessary to guarantee the correct versioning and distribution of updated libraries and models. Open-source libraries may change, deprecate, and update over time, leading to incompatible versions of kernels, architectures, libraries, dependencies, and application program interfaces (APIs), which need to be constantly monitored and documented. Another limitation of open-source software is that they do not have a warranty for misuse, commercial untrained utilization, responsibility disclaims, versioning (forking), or any other support or legal constraints associated with proprietary software. As it concerns modeling accuracy, it is expected that the solution of linearized optimization problems may lead to simulation results that differ from the real plant conditions. However, those approaches have proven to produce sufficiently reliable results in various industrial sectors [41,42]. Finally, execution errors are typically expected for computational tools that orchestrate different open-source libraries and suites. It entails a compromise between the versatility, shareability, and usability of the tool, although the authors envisage rather an unlocked potential advantage for making scientific work more automatic, integrated, and reproducible by centralizing the energy audit data, the mathematical modeling, and the reporting capabilities.

2. Methods

In this section, the structure, workflow, and functionalities of the ROSMOSE tool are detailed, such as the connection strategy to other software for thermodynamic properties calculation, use of open-source optimization solvers, plotting and visualization libraries, and bibliography data management. A case study that demonstrates the application of the ROSMOSE tool is stated, based on a conventional dairy process, its auxiliaries, and waste upgrading energy conversion systems. The key performance indicators adopted for comparing the several integration options for the dairy plant are also summarized.

2.1. Description of the conventional dairy process

In a typical dairy plant, raw milk is first transformed into concentrated and pasteurized milk. Next, it is converted into value-added products, such as cream and cheese. The overall system comprises multiple processing sections in which the operating temperatures and flowrates are defined by the processing recipes. A general block flow diagram for the dairy plant, with the feedstock, support, product, and waste streams, is depicted in Fig. 1.

Fresh milk is received, cooled, and stored. Next, it is heated up to 60 °C and separated from the cream via centrifugation. The milk stream is further heated to 75 °C and quickly cooled to 4 °C in pasteurization to destroy harmful bacteria and ensure sufficient shelf-life. The cream is heated up to 70 °C and next sent to a mixer, where the thickener is added, and the mixture is gradually cooled down to the storage temperature of 4 °C. Meanwhile, the pasteurized milk is sent to the evaporation section, wherein the milk is preheated and passed through a series of multi-effect evaporators. In the first evaporator, saturated steam is used to evaporate part of the water in the milk. The produced steam is used to preheat the inlet milk and evaporate more water from the concentrated milk in the second effect and similarly in the third effect. In order to allow for such heat recovery, the pressure in each evaporator is lowered by throttling the milk flow from the pressure of the preceding one. This approach does not recover all the evaporation enthalpy from the produced vapor, as there is usually more vapor production from the first effect than the second effect evaporator can use [43,44]. Next, the downstream cheese production section involves preheating the concentrated milk up to 56 °C, adding cheese-forming cultures, and filtering out the whey (or lactoserum) [45]. Further cheese processing involves cooling and the addition of water and salt. The auxiliary energy systems necessary for the operation of the dairy plant include cleaning in place, storage, and wastewater treatment units, which are important to consider in the overall energy balance of the dairy production site.

Fig. 2 shows a more detailed process flow diagram including typical units of a dairy plant, together with the utility system and possible integration routes to produce value-added by-products. Alternative processes are suggested for valorizing the whey streams separated in the evaporation and cheese production sections. The first option involves the production of brown cheese, known as mesost, which involves evaporating part of the water content in the whey, mixing it with cream, and cooling it down to the storage temperature. The second alternative

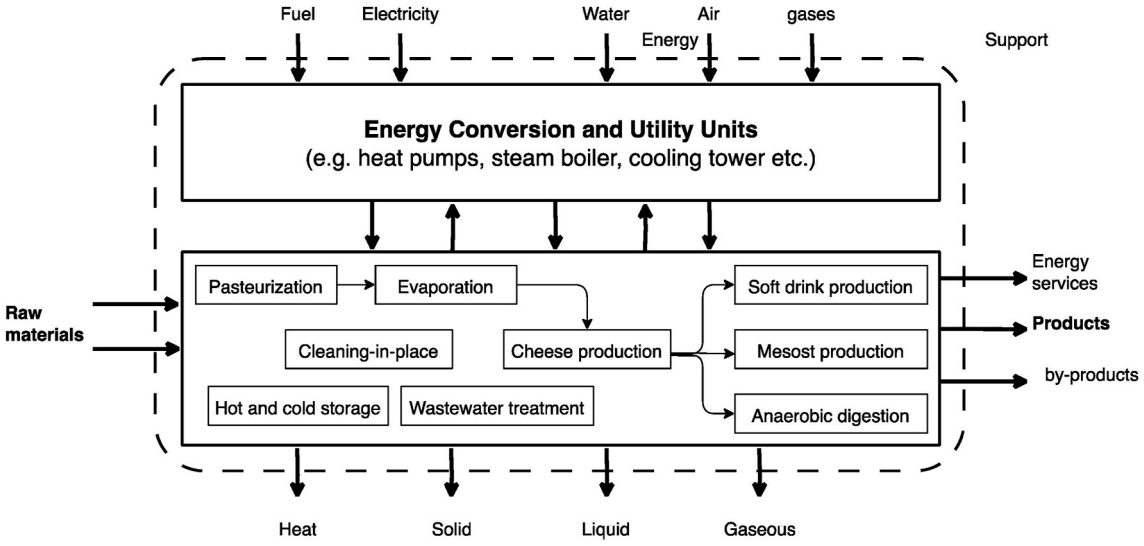


Fig. 1. Dairy plant block flow diagram, showing the feedstock, support, product and waste streams, and the battery limits.

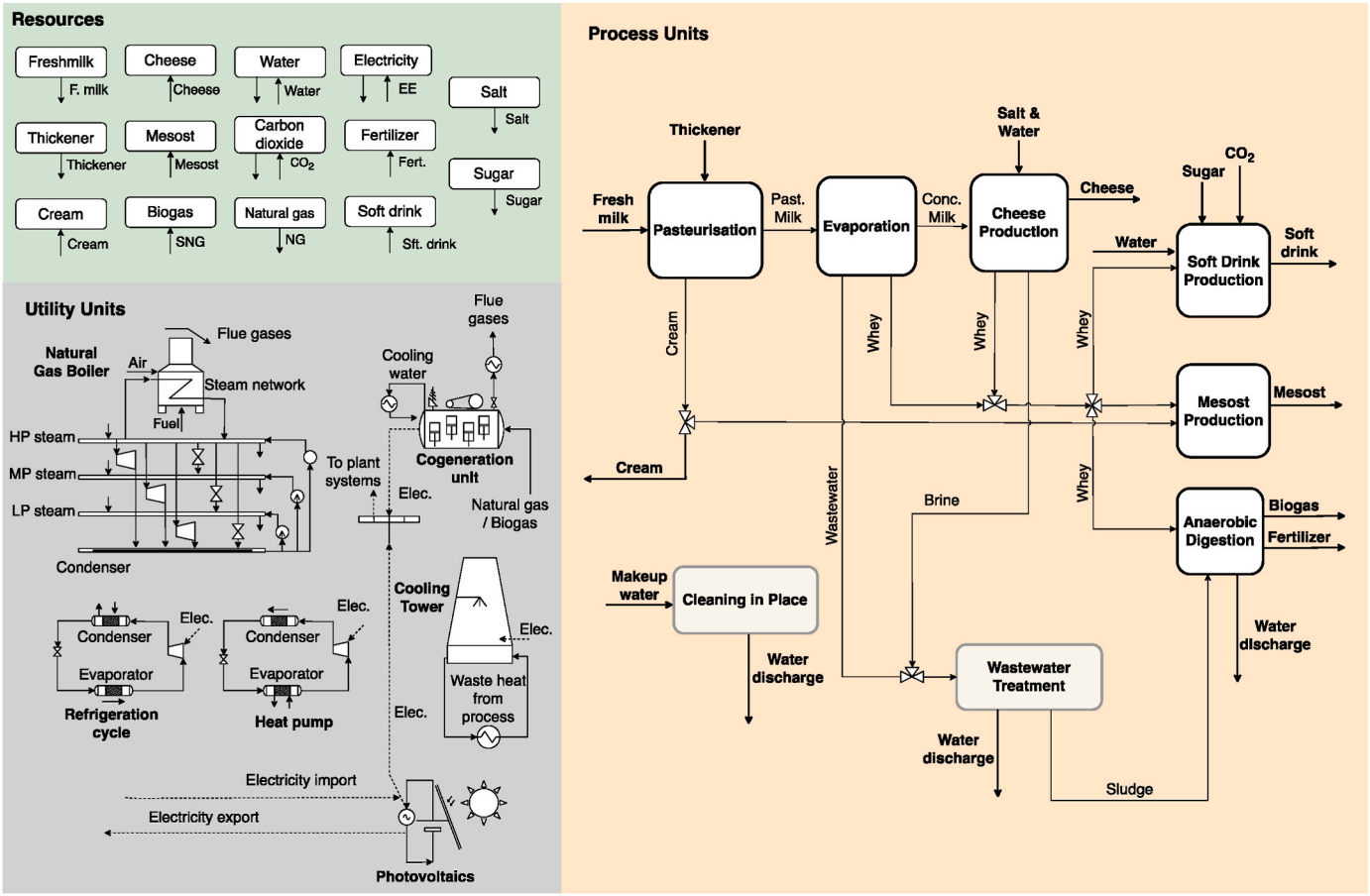


Fig. 2. Detailed flow diagram of a dairy plant, with the resource streams, and utility integration options.

utilizes the whey stream for producing a whey-based soft drink. It involves evaporating a significant portion of the water in the whey stream, cooling the concentrated whey to 50 °C, mixing fruit sugar and flavoring ingredients, and finally adding water and CO2 to produce the carbonated soft drink [46].

The third and final alternative uses the organic content of the whey and sludge streams to produce biogas (55 % CH_4, 45 % CO_2) through anaerobic digestion in a biodigestion reactor operating under thermophilic conditions [47–49]. Heating is needed to maintain the reaction temperature in the digester. Steam is used to dry the dewatered digestate that can be sold as solid fertilizer due to its high organic load [50]. The biogas can be burnt onsite to supply combined heat and power in a cogeneration unit or upgraded into biomethane for injection into the local natural gas grid. As an example, Germany currently operates more than 9000 biogas plants, which provided about 50 TWh of the domestic energy demand (electricity, heat, or gas) in 2021 [51]. Like any other gas processing facility, operational constraints must be put in place to ensure the safe and efficient operation of the plant, including factors such as controlled temperature, retention time, and composition within the bioreactor [52]. Another alternative for supplying renewable energy to the overall system is the integration of photovoltaic (PV) panels to harness solar energy. In a study by Murali et al. [53], a solar photovoltaic thermal (PVT) system was integrated into a livestock farm to evaluate the potential for replacing fossil fuel utilization. The electricity generated was used to operate heat pumps and ancillary plant requirements, while the thermal energy was stored in an integrated storage system. Such an alternative showed promising results in avoiding approximately 20 tonnes of CO_2 emissions per year and will be considered among the decarbonization options for the system under evaluation in this work.

Accordingly, multiple integration opportunities arise, and the operating parameters of the energy technologies strongly influence the plantwide mass and energy balances. Hence, a systematic approach must be adopted to evaluate the performance of various integration options for the dairy case study. To this end, the ROSMOSE tool is used to perform the energy integration and total site optimization of the conventional dairy process and to graphically represent the key performance indicators described in the following sections.

2.2. *ROSMOSE structure and workflow*

The ROSMOSE platform is based on the OSMOSE engine and the Quarto environment. OSMOSE is a process integration tool developed at the Industrial Processes and Energy Systems Engineering (IPESE) group from École Polytechnique Fédérale de Lausanne (EPFL) in Switzerland. The acronym stands for “Optimization Multi-Objectifs des Systèmes Énergétiques Intégrés” (in English, Multi-Objective Optimization of Integrated Energy Systems). It has been used for over two decades in research projects and teaching activities in EPFL and undergoes continuous development and the addition of new features [54–58]. As a computational platform, OSMOSE utilizes a combination of metaheuristic (genetic algorithm, GA) and deterministic (Mixed Integer Linear Programming, MILP) optimization approaches for combined heat and mass integration. This framework aids in decision-making for the design and operation of energy systems, either industrial or urban, by considering thermodynamic, environmental, and economic targets. Moreover, the OSMOSE platform exhibits flexibility in integrating various external software and tools, like database handling routines (CSV and JSON files), process modeling and simulation suites (Aspen Plus, Belsim Vali, Coolprop, DWSIM), optimization solvers (AMPL,

Pyomo), data visualization (GNU Plot, D3.js, Mermaid, Matplotlib).

ROSMOSE tool was developed to simplify the utilization of the complex OSMOSE language, which operates in the backend, making it more accessible to a wider scientific community. ROSMOSE also extends OSMOSE functionalities by including a complete reporting suite built in the Quarto ecosystem, capable of rendering Quarto (.qmd) and Rmarkdown (.Rmd) formats and directly converting them into a report or a presentation. The aim is to make science reproducible by creating an ecosystem where programming can be embedded into reports [36,59,60]. Compared to conventional reporting tools, in which data is stored as characters and figures are deployed as static images, a Quarto-generated report includes in the same document the explanatory text, raw data, modeling parameters, and plotting instructions to generate the figures. The report is built based on the raw data and the executable routines embedded in the code chunks, which can automatically handle the figures updating as soon as the data undergoes any

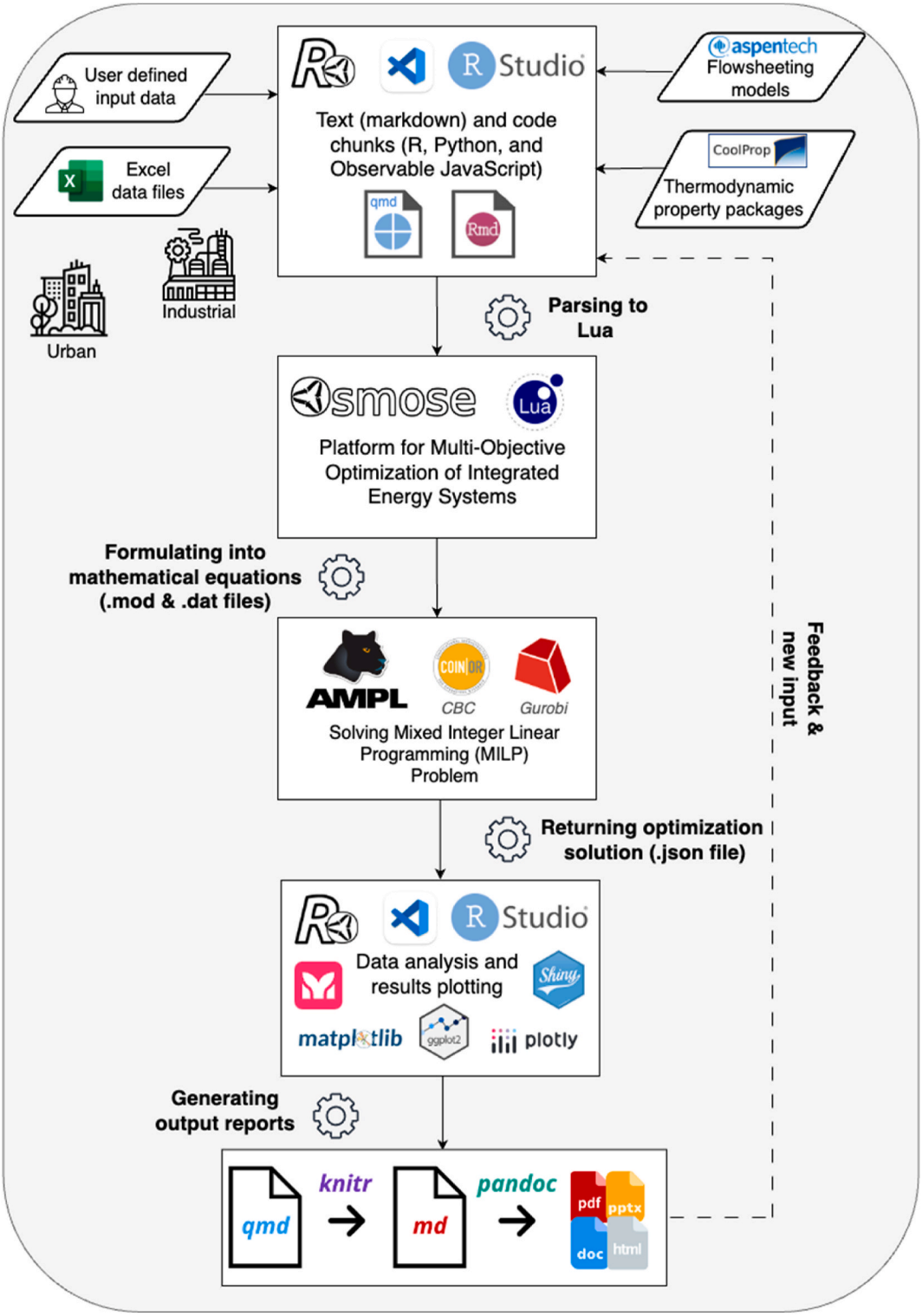


Fig. 3. Schematic illustration of the working mechanism of ROSMOSE.

changes.

Fig. 3 illustrates the workflow of the ROSMOSE tool, highlighting the mechanisms of data transfer and interactions between various platforms, starting from the model parametrization through the results visualization and finally the report generation in multiple output formats. It is therefore worth highlighting that the ROSMOSE tool is not an alternative problem-solving algorithm for energy systems integration but rather a powerful platform that combines multiple functionalities for software integration, open-source and commercial mathematical solvers execution, data and metadata management, and more importantly, documenting, reporting, and visualization methods. This high-level programming and reporting tool relies on familiar tabular and graphical solutions for synthesizing and optimizing complex industrial plants. Moreover, ROSMOSE has already been tested and validated by industrial experts in food processing industries, tanneries, distilleries, and breweries, aiming to integrate heat pump solutions for decarbonization purposes. Some features, such as parsing and importing structured data from Excel files, have been adopted to ease inputting information to ROSMOSE by industrial facilities [61–65].

Multiple programming languages can be used in code chunks in a .Qmd document, including Python, R, Julia, and Observable JavaScript. For example, Python could be used along with the open-source library Coolprop to calculate the thermophysical properties of various substances involved in systems like the milk processing plant, as well as in rigorous heat pump or steam network superstructures [37]. For instance, to calculate the thermodynamic state that defines each of the processing streams in the dairy plant, two Python functions have been programmed, namely the State and Mixture functions. The former function requests two independent intensive thermodynamic properties, such as temperature and pressure (or pressure and vapor quality), and returns the remaining thermodynamic properties that fully define the state, such as enthalpy, entropy, heat capacity, etc. On the other hand, the Mixture function allows to estimate the properties of the milk-related streams as a mixture of water and fat, the proportions of which vary according to the processing stage within the dairy plant. Coolprop has an extensive database of more than 100 fluids, and their properties can be accessed via ROSMOSE using the Python wrapper. Alternatively, thermodynamic properties of other substances can be imported from chemical flowsheeting software, such as Aspen Plus or DWSIM. The features of Python language and Coolprop library could be boosted by using other programming libraries, such as NumPy and SciPy, to solve systems of equations that are associated with industrial mass and energy balances, vapor-liquid equilibrium calculations, design specifications or sizing routines. It is worth noticing that ROSMOSE leverages the concepts of blueprints for developing industry models. These blueprints consist of a series of ex-ante models providing insights into the key inputs and outputs of a given industry in terms of thermal and electrical energy, materials, and services. A blueprint is not necessarily a detailed model of process operations, although it can be done as detailed as desired. Thus, it is typically unsuitable for advanced process supervision or control and is not fitted to study dynamic systems. The validity domain is restricted to nominal operations and stationary regimes. The time resolution of a certain optimization problem may be defined from a yearly or monthly scale up to one of hourly or minute details.

In ROSMOSE, any variable can be defined as a *tag*, which transcends a simple numeric value. In fact, a *tag* is an object containing a name, a value, a physical unit, and a description. The values of the tags can be either user-defined or calculated based on the mathematical relations among other tags. Tags specify the inputs, outputs, or parameters of a specific model. The building blocks of a ROSMOSE project are the *energy technology (ET)* models. Each ET can have multiple input and/or output tags, as well as other attributes, such as *layers* and *units*. The layers in ROSMOSE can be interpreted as the pipelines through which similar streams (or information) flow and are defined by a name and a physical unit. Moreover, a bibliography data management package has been incorporated into the ROMSOSE reporting environment in Quarto to offer an automatic method for referencing and documenting the parameters and metadata used to generate the results. This is done by linking the report to reference management systems, such as Zotero, which associate tags to the references. Fig. 4 provides an example of a ROSMOSE project structure and typical relationships between ETs, units, and layers. The units belonging to a specific ET can be of two types, either *process* or *utility* units. The main difference between them is that, on the one hand, a *process* unit corresponds to an industrial production process, whose requirements must be satisfied and, therefore, its sizing factor should be fixed during the optimization problem. On the other hand, the *utility* units represent auxiliary equipment whose size is optimized to meet the process energy requirements. For instance, a process unit can be defined based on the operating data of a dairy plant (e.g., the resource flows, the supply and target temperatures, the composition, the heating demands, the specific costs, the process constraints, and scheduling, etc.). In contrast, the set of potentially integrated utility units, such as heat pumps, steam boilers, cooling systems, refrigeration units, cogeneration units, and grids, can be resized to meet the dairy process demands. The units, either processes or utilities, have multiple attributes including.

i) specific operating (EUR/$kg_{product}$ or EUR/$kWh_{product}$) and investment costs (EUR/$(kg/h)_{product}$ or EUR/$kW_{product}$),
ii) environmental impact factors ($kg_{CO2}/kg_{product}$ or $kg_{CO2}/kWh_{product}$)
iii) size factors (continuous optimization variables in the case of utility units, and fixed in case of process units), and
iv) activation factors (binary optimization variables).

Moreover, depending on the main purposes of the energy conversion processes (heating, cogeneration, power production, waste treatment, mechanical processing, separation, storage, etc.), two types of streams may be assigned to each unit:

- Resource streams, which are linked to one of the previously defined layers and require an indication of their direction regarding the unit, namely *in* (for consumption) or *out* (for production).
- Heat streams, which correspond to the thermal energy flows and require information such as inlet and outlet temperature, inlet and outlet enthalpy (also heat capacity or heat flowrate), minimum temperature approach, and heat transfer coefficient.

2.3. Flowsheeting software connection and modeling of heat pump and steam network superstructures

Industrial processes involve different types of substances and energy conversion systems, which require the use of specialized databases and mathematical solvers for calculating the thermophysical properties and the energy performance of the equipment. To this end, different software can be integrated into the ROSMOSE tool to determine the thermodynamic states and perform the mass and energy balances. Results are retrieved from external software, properly linearized, and used in the MILP optimization problem of energy integration. For instance, according to Fig. 5, process specifications can be defined in a flowsheeting software, such as Aspen® Plus or DWSIM, and resulting energy and mass flows retrieved into ROSMOSE, such as the power consumption in the compression of CO_2 up to the supercritical state for subsequent injection purposes. A similar connection can be implemented for the DSWIM open-source process simulator [66,67].

Alternatively, a Python wrapper of the Coolprop library can be used in an equation-oriented (EO) modeling approach to simulate two main superstructures of utility systems as those shown in Fig. 6. The first superstructure incorporates a detailed mass and energy balance of a steam network equipped with a Rankine cycle of multiple pressure levels [69]. The second superstructure consists of a multistage heat pump with a saturator, economizers, and subcooling options [70]. The pressure

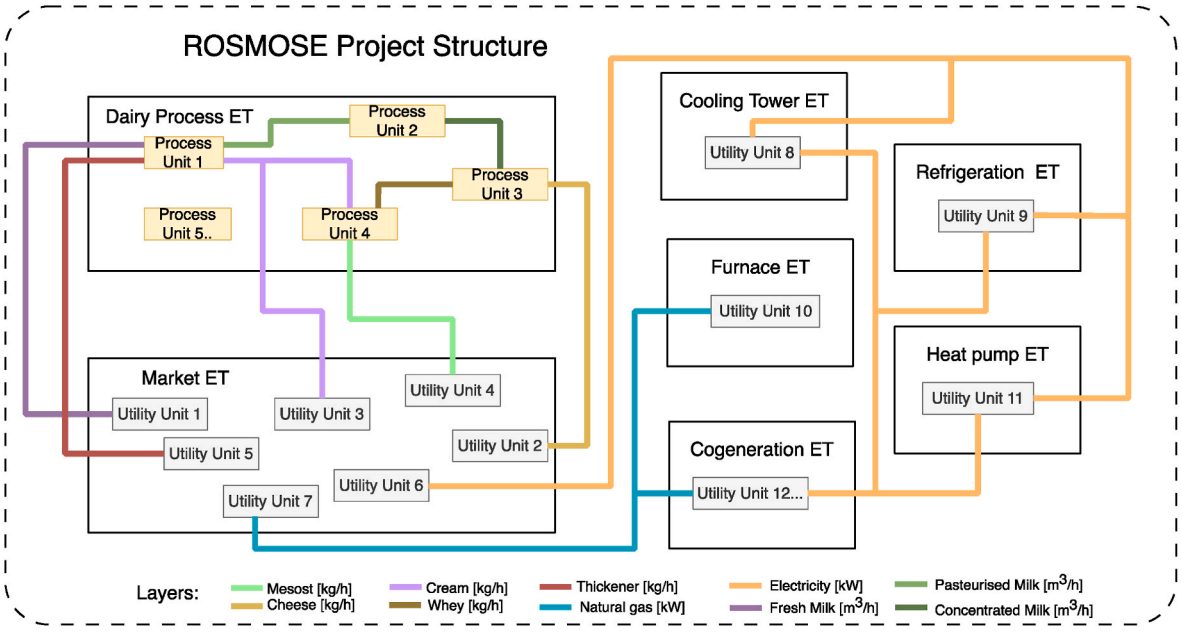


Fig. 4. ROSMOSE project structure and the relationship between ETs, units, and layers pertaining to the dairy plant sample project.

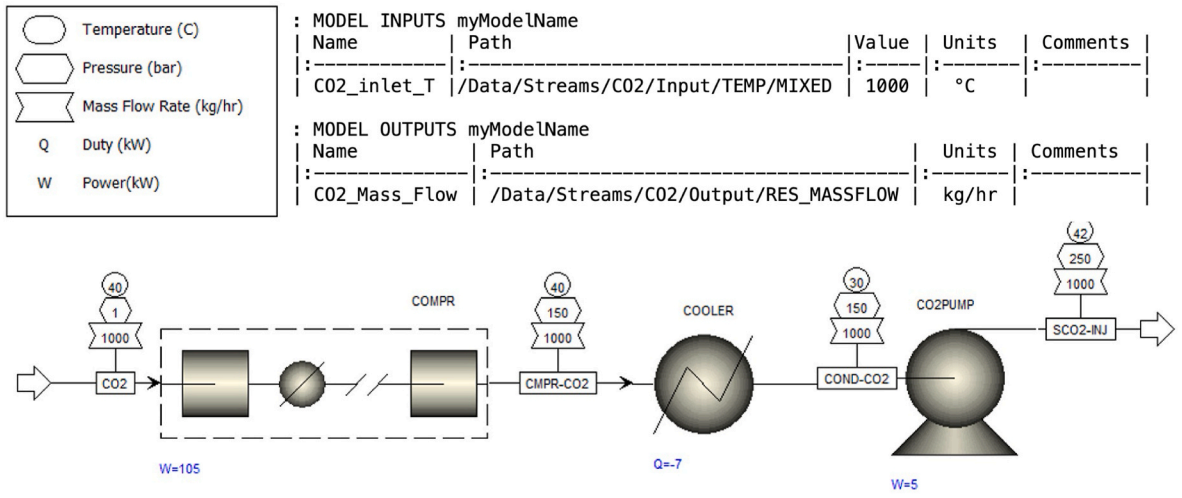


Fig. 5. Aspen® flowsheet example of CO_2 injection unit in ROSMOSE tool [68].

levels of each superstructure are set by the user based on the examination of the grand composite curve of the dairy or other industrial processes. The user can also list the fluids among which one or more can be chosen by the optimization problem solved by the ROSMOSE tool. This approach has been successfully applied in other works [71,72]. More recent studies aim to extend the capabilities of the ROSMOSE tool by using a data-driven systematic methodology for the prediction of optimal heat pump integration using machine learning [73].

2.4. Optimization problem definition

For an industrial process, multiple competing utility setups and operating conditions entail many energy integration scenarios. For example, when utilizing a cogeneration utility unit in a dairy process, there is a possibility of importing fossil fuel or using biogas produced locally in a digester to satisfy part of the heating requirements of the system. Another alternative could be to import electricity to avoid the inefficiencies of a cogeneration unit used to supply the power and heat demands of the ancillary technologies, such as refrigeration, cooling water, and heat pumps (Fig. 2). ROSMOSE tool can be used to calculate the minimum energy requirements (MER) of the process by applying pinch analysis techniques. Next, it can provide the best combination(s) of energy technologies and their corresponding operating conditions that reduce the total cost. To this end, the MILP problem shown in Eq. (1) is handled by ROMOSE and solved by the OSMOSE engine and AMPL suite operating in the backend [68].

$$min\ Cost_{tot} = \sum_{u \in Units} \left[\sum_{t \in Time} \left(Cop1_u \bullet Y_{u,t} + Cop2_u \bullet M_{u,t} \right) \bullet \Delta t + Cinv1_u \bullet Y_u + Cinv2_u \bullet M_u \right] \quad (1)$$

where.

- $Cost_{tot}$ is the total annual cost of the industrial energy system [€/y],
- $Cop1_u$ is the fixed operating cost of unit u [€/hr],

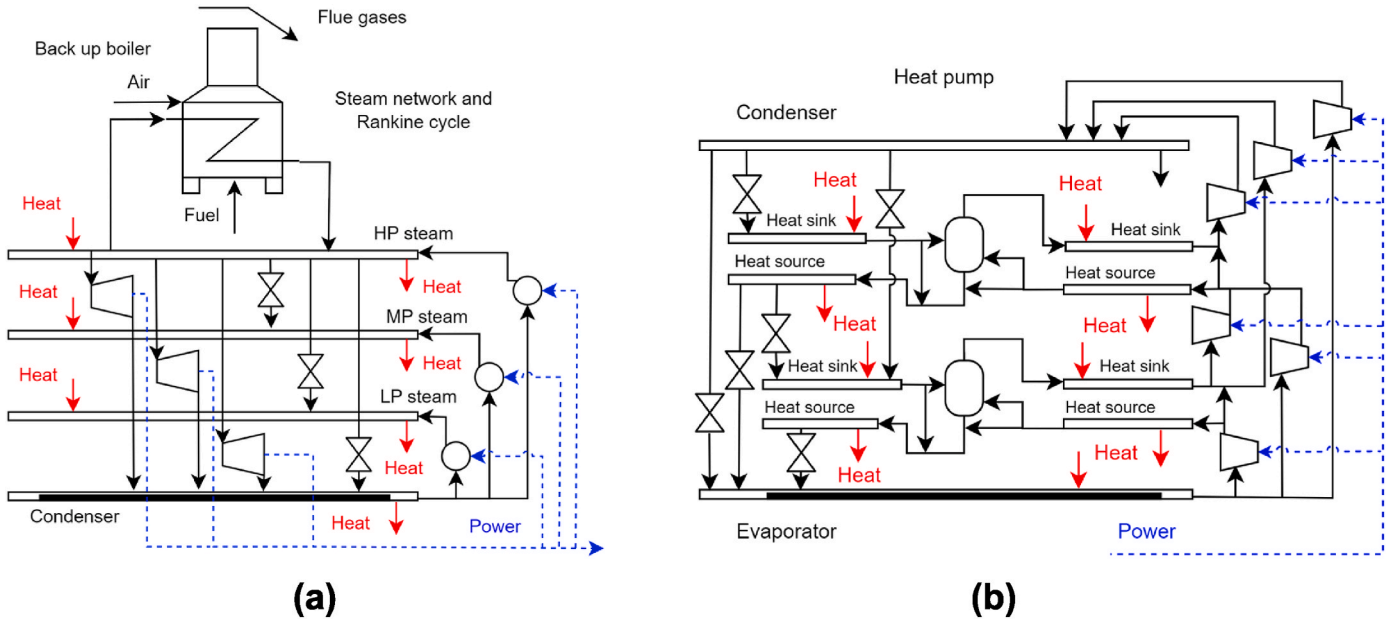


Fig. 6. Superstructures of (a) heat pump and (b) steam superstructure available in ROSMOSE tool [74].

- $Y_{u,t}$ is the integer variable related to the presence or absence of each unit u at time t [−],
- $M_{u,t}$ is the corresponding continuous load of a unit u at time t [kg/hr or kW],
- $Cop2_u$ is the variable operating cost of unit u [€/kg or €/kWh],
- Δt is the annual operating time of unit u [hr/y],
- $Cinv1_u$ is the fixed annualized investment cost of unit u [€/y],
- Y_u is an integer variable related to the presence or absence of a unit u in the system; the maximum value of $Y_{u,t}$ [−],
- $Cinv2_u$ is the specific annualized investment cost of a unit u [€/kg or €/kWh],
- M_u is the corresponding maximum load of the unit u [kg/y or kWh/y].

This objective function is subject to multiple constraints including unit existence and load boundary conditions:

$$Y_{u,t} \bullet Fmin_u \leq M_{u,t} \leq Y_{u,t} \bullet Fmax_u \;\; ; \quad \forall u \in Units, \forall t \in Time \tag{2}$$

$$M_{u,t} \leq M_u \tag{3}$$

$$Y_{u,t} \leq Y_u \tag{4}$$

Mass and energy balance constraints:

$$\sum m_{Input\ streams} = \sum m_{Output\ streams}; \quad \forall Layer, \forall Time \tag{5}$$

Power production or consumption constraints:

$$\sum_{w=1}^{N_w} M_w \bullet w_{w,utilities} + \sum_{Processes} W + W_{import} - W_{export} = 0 \quad \forall Time \tag{6}$$

Heat cascade constraints, balance at the temperature interval r:

$$\sum_{w=1}^{N_w} M_w \bullet q_{w,r} + \sum_{i=1}^{N} Q_{i,r} + R_{r+1} - R_r = 0 \; ; \quad \forall r = 1.....N \tag{7}$$

And solution feasibility:

$$R_1 = 0, R_{N+1} = 0, R_r \geq 0 \quad ; \quad W_{import} \geq 0, W_{export} \geq 0 \quad \forall r = 1.....N \tag{8}$$

where.

- N is the number of temperature intervals based on the supply and target temperatures of all streams [−]
- Q is the heat exchanged between process streams [kW],
- R is the heat cascaded from higher (r+1) to lower (r) intervals of temperature [kW],
- N_w is the number of utility units available,
- $q_{w,r}$ is the specific heating or cooling flow supplied by any utility unit w within an interval r [kWheat/kW or kWheat/kg],
- W is the power produced or consumed by the utility systems, the unit processes, or imported from or exported to the grid [kW].

Additional thermodynamic and environmental indicators can be determined to rationally compare and represent scenarios in non-dominated Pareto plots. A weighted combination of multiple objective functions, such as costs and impact factors, may also be used to govern the optimization problem. ROSMOSE triggers the execution of the routines in the OSMOSE engine, which writes the respective *.run*, *.mod,* and *.dat* files in AMPL language. Those files contain the equations of the optimization problem and the parameters of the ET models. The mathematical formulation of the problem can be solved using any open-source MILP, such as CBC [75], which is compatible with the AMPL® community edition. Once the optimization problem is solved, results are dumped into a solver output file (JSON format) and parsed to extract and structure the data. Post-compute functions are next used to.

- Calculate performance indicators (e.g., costs, environmental impact, energy efficiencies, etc.).
- Display graphical representations of relevant results (e.g., composite and integrated curves).
- Calculate objective functions for multi-objective optimization scenarios.
- Carry out post-optimization analyses and issue recommendations.

In the case study of this paper, the costs of all input and output streams in the dairy plant are assumed according to the current European market and summarized in Table 2.

2.5. Key performance indicators (KPIs)

Thermodynamic, economic, and environmental performance indicators are proposed to compare the different process alternatives.

Table 2
Material and utility costs assumed in the dairy process case study.

Material/Utility stream	*Unit*	*Cost*
Water	€/m3	2.50
Fresh milk	€/m^3	500
Electricity	€/kWh	0.25
Natural gas	€/kWh	0.12
Biogas	€/kWh	0.08
[a]CO_2 tax	€/tonne	100
[b]CO_2	€/kg	0.008
[c]Thickener	€/kg	0.87
Salt	€/kg	0.13
Sugar	€/kg	0.64
Cream	€/kg	2.21
Cheese	€/kg	8.78
Mesost *(brown cheese)*	€/kg	20.0
Soft drink	€/L	2.50
Fertilizer	€/kg	0.90

[a] Value based on typical CO_2 tax range in European countries, Switzerland being on the high end.
[b] Cost of pure CO_2 materials stream needed to produce soft drinks.
[c] Assumed to be corn starch.

Firstly, the energy consumption per unit volume of fresh milk processed in the dairy system (φ) is calculated to evaluate the overall energy efficiency of the system. The metric is divided into fuel consumption efficiency φ_{fuel} and electricity consumption efficiency φ_{EE}, computed based on Eq. (9) and Eq. (10), respectively.

$$\varphi_{fuel}\left[\frac{kWh_{fuel}}{m^3_{freshmilk}}\right] = \frac{Q_{NG,furnace} + Q_{cogen}}{\dot{V}_{freshmilk}} \tag{9}$$

$$\varphi_{EE}\left[\frac{kWh_{EE}}{m^3_{freshmilk}}\right] = \frac{W_{refrigeration} + W_{CoolingTower} + W_{heat\ pumps}}{\dot{V}_{freshmilk}} \tag{10}$$

where.

- W is the power consumption in the utility units (refrigeration, cooling tower, and heat pumps) [kW].
- Q is the fuel consumption of the natural gas furnace and the cogeneration unit [kW],
- $\dot{V}_{freshmilk}$ is the volumetric flowrate of fresh milk in [m3/hr] which (~28 m^3/h for this case study).

The total CO_2 emissions, encompassing both direct and indirect emissions, are calculated per m^3 of processed milk according to Eq. (11).

$$CO_{2,Spec}\left[\frac{kg_{CO_2}}{m^3_{freshmilk}}\right] = \left[Q_{NG,total} \bullet 3600 \bullet \left(\frac{I_{CO2,NG}}{LHV_{fuel}} + \frac{r_{CO2,NG}}{1000}\right) + \frac{r_{CO2,EE} \bullet W_{total}}{1000}\right] \bullet \frac{1}{\dot{V}_{freshmilk}} \tag{11}$$

where.

- $I_{CO2,NG}$ is the direct emissions factor for natural gas considered as 2.75 [$kgCO_2/kg_{fuel}$],
- $r_{CO2,EE}$ is the indirect emissions factor of electricity 62.63 [gCO_2/kWh] [76],
- $r_{CO2,NG}$ is that of natural gas 0.0049 [gCO_2/kJ] [76],
- LHV_{fuel} is the lower heating value of the fuel considered for natural gas as 50,000 [kJ/kg],
- $Q_{NG,total}$ is the total natural gas burning load [kW],
- W_{total} is the total electric power consumption in the system [kW].

Finally, the total revenue is calculated by discounting the incremental annualized investment costs of the utility systems and the total annualized operating costs from the annualized revenues of the product streams, as shown in Eq. (12):

$$Revenue\left[\frac{€}{m^3_{freshmilk}}\right] = \left[\sum_{i\in products}\left(M_{product,i} \bullet C_{product,i}\right) + M_{AddedValue} \bullet C_{AddedValue} \pm W_{total} \bullet C_{electricity} - Q_{NG,total} \bullet c_{NG} - \sum_{u\in Units} Z_u \bullet F_{annual,u} \bullet y \Big/ 8760h\right] \bullet \frac{1}{\dot{V}_{freshmilk}} \tag{12}$$

where.

- C_i is the cost of flow i per unit of dairy product, feedstock, or commodity [€/kg or €/m^3 or €/kWh],
- $M_{byproduct} \bullet C_{byproduct} = (M_{mesost} \bullet C_{mesost})$, for the scenario of mesost production [€/h],
- $M_{byproduct} \bullet C_{byproduct} = (M_{soft\ drink} \bullet C_{soft\ drink})$, for the scenario of soft drink production [€/h],
- $M_{byproduct} \bullet C_{byproduct} = (M_{biogas} \bullet C_{biogas} + M_{fertilizer} \bullet C_{fertilizer})$, in case of anaerobic biodigestion [€/h],
- Z_u is the investment cost of utility unit u [€],
- $F_{annual,u}$ is the annualization factor based on the lifetime and the annual interest rate (~7 %).

3. Results and discussion

The performance of a conventional dairy processing facility is compared to alternative routes in which *(i)* waste streams are upgraded into value-added products and *(ii)* optimized utility systems are integrated to minimize the overall cost and exergy losses. The minimum energy requirements of the process are reported, and the main energy consumption sources are identified. In addition, Carnot composite curves of the base case and the optimal integrated scenarios are depicted for all process routes. These curves allow for a better understanding of the integration of the utility systems and highlight the sources of inefficiencies in each setup. Finally, thermodynamic, environmental, and economic indicators are used to rank the different configurations, considering carbon taxation and costs of energy inputs. ROSMOSE returns these optimization results in a.json output file and displays a customizable .html report.

3.1. Energy consumption remarks and process integration results

As a food processing site, the dairy process does not inherently contain any endothermic reactions, but rather a series of heating and cooling steps, with some heat dissipation introduced due to power consumption in certain equipment (e.g., centrifuge, compression, pumping, stirring, etc.). The minimum energy requirements of the dairy plant producing mesost cheese are determined and graphically represented in the composite curves shown in Fig. 7 (a) and (b). The total energy integration potential between process-to-process streams is approximately 391 $kWh/m^3_{freshmilk}$ (~10,930 kW), which can be attained by installing a HEN with an exchanging area of approx. 3000 m^2 and a total annualized cost of roughly 1.38 €/$m^3_{freshmilk}$ (~338,800 €/year). For the mesost production scenario, the minimum energy requirements are estimated as 125 $kWh/m^3_{freshmilk}$ of heating utility (i.e., 3,485 kW) and 114 $kWh/m^3_{freshmilk}$ of cooling utility (i.e., 3,200 kW), which is more clearly appreciated in the grand composite curve of the process shown in Fig. 7 (b). Similar trends in heating and cooling utility requirements are found for other process configurations producing soft drinks (Fig. 8 (a) and (b)) or biogas (Fig. 9 (a) and (b)).

As summarized in Table 3, a reduction of 75 %–90 % in total fuel consumption is expected for all the value-added production routes that

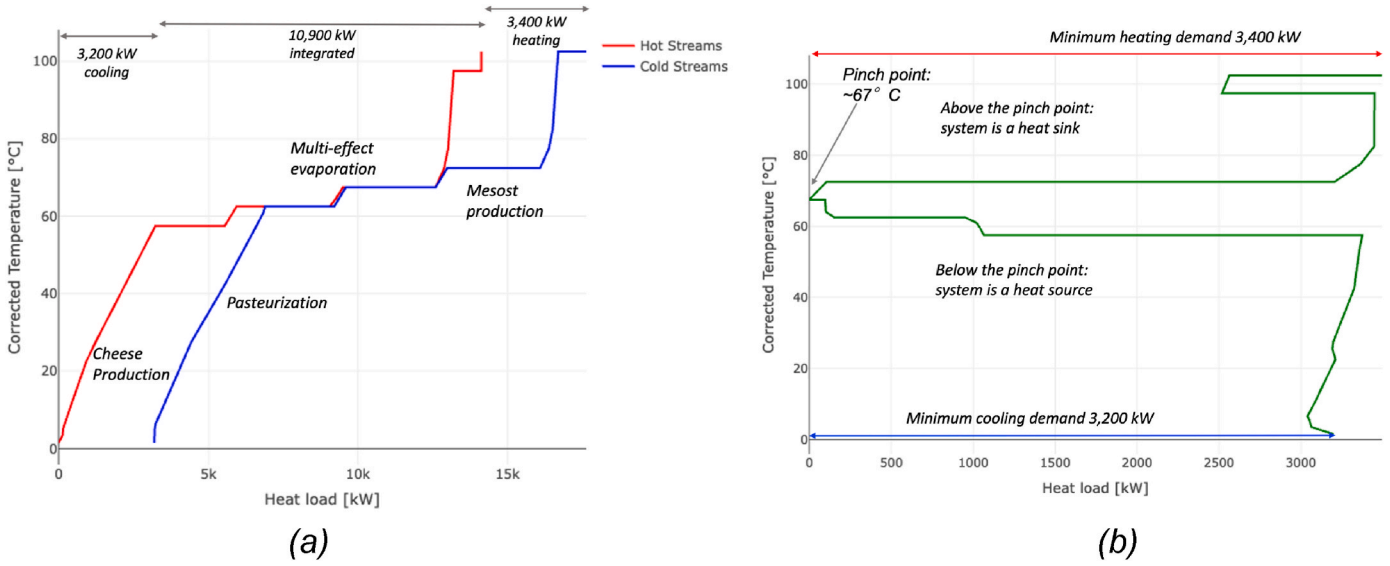


Fig. 7. Pinch analysis curves of a dairy plant (mesost production base case) a) Hot and cold composite curves. b) Grand composite curve, as generated in ROSMOSE report.

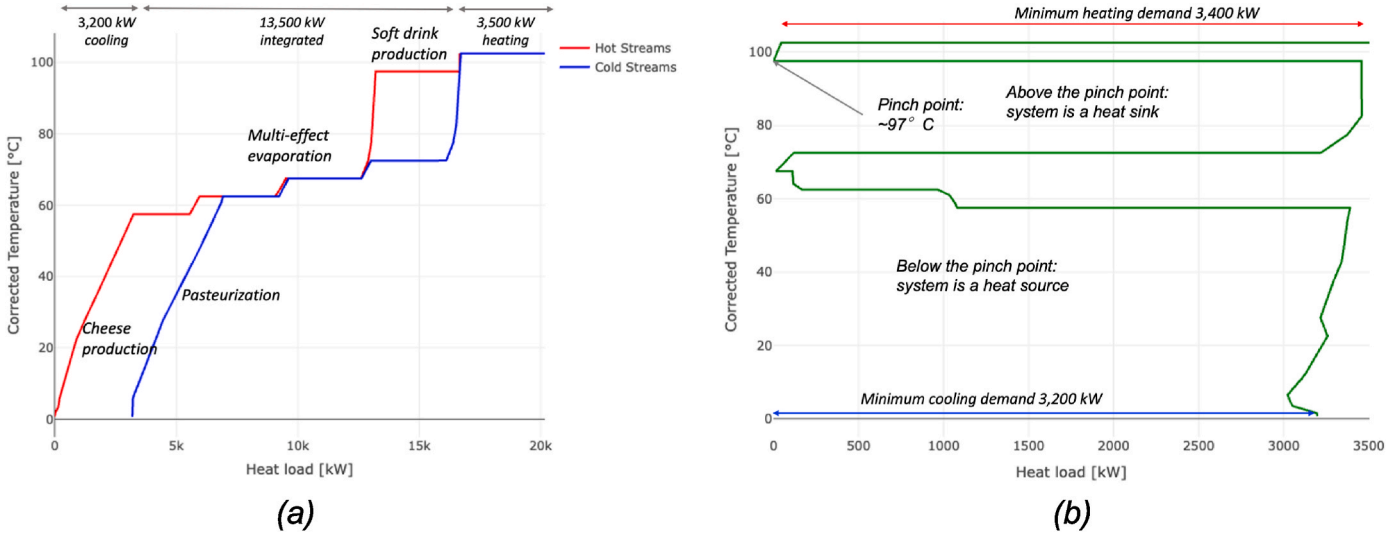


Fig. 8. Pinch analysis curves of a dairy plant (soft drink production base case) a) Hot and cold composite curves. b) Grand composite curve, as generated in ROSMOSE report.

integrate heat pumps for supplying the process heating demands. For example, in the mesost production route, the overall site fuel consumption may be reduced from 134 to 30 kWh/m^3. This gain comes at the expense of an additional electricity consumption of 3–4 times for the various scenarios. To tackle this issue, an internal cogeneration unit can supply between 35 % and 100 % of the additional power demands of the plant. All in all, the advanced utility scheme demonstrates a strong interest in achieving an optimal combination of waste heat recovery below the pinch point temperature and its valorization above that point to partially meet the heating requirements of the industrial system. In addition to the fuel and power requirements, Table 3 also compares the overall product yield of the three routes for converting the whey stream (a by-product initially considered as waste in the dairy plant) into value-added products, such as mesost cheese, soft drinks, and biogas. Due to a better overall yield of the mesost production route per m^3 of whey, this option generates the largest product amount per unit volume of processed feed.

Fig. 10 (a–b) depicts a graphical representation of selected utility integration options for the dairy plant in the mesost production scenario. The elimination of excessive steam generation using natural gas boilers is enabled by integrating heat pumps and achieving optimal waste heat recovery below the pinch point temperature. This utility optimization brings about the reduction of the area between the utilities and process integrated Carnot plots (ICP). In this type of diagram, the main y-axis is the Carnot factor (1-T_0/T), whereas the grey-shaded area corresponds to the exergy destruction in the HEN [77]. The smaller the area between the curves, the lower the exergy destruction or, equivalently, the lower the entropy generation in the dairy production site. This fact is evident when comparing Fig. 10 (a) using conventional utilities (i.e., large grey shaded area) and Fig. 10 (b) using heat pumps and a cogeneration system. A similar conclusion is drawn for the soft drink and biogas production routes, where using an optimal combination of a cogeneration engine and heat pumps significantly reduced the thermal exergy destruction of the HEN. This is graphically represented in the reduction of grey shaded area for Fig. 11 (b) compared to (a) in the soft drink production case and, similarly, for Fig. 12 (b) compared to (a) in the

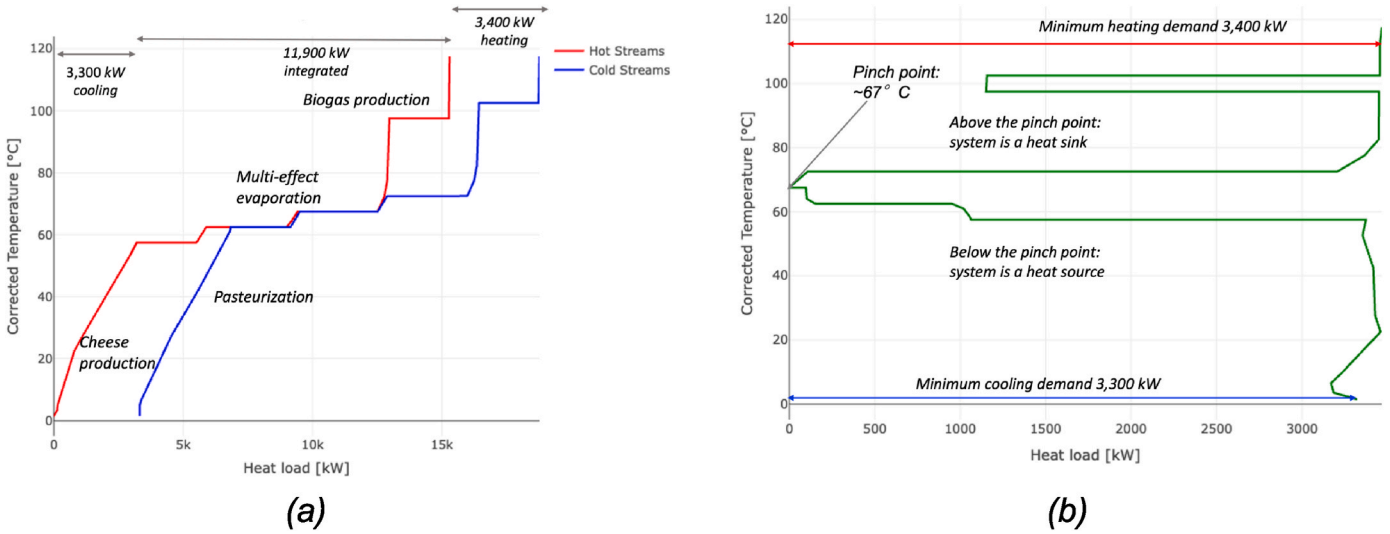


Fig. 9. Pinch analysis curves of a dairy plant (biogas production base case) a) Hot and cold composite curves. b) Grand composite curve, as generated in ROSMOSE report.

Table 3
Energy consumption remarks of the dairy plant and other value-added production routes depending on the energy integration of heat pumps and cogeneration systems.

Process layout	Mesost production (assumed base case)		Soft drink production		Biogas production	
	Conventional	Heat pump & Cogeneration	Conventional	Heat pump & Cogeneration	Conventional	Heat pump & Cogeneration
Total electricity import from grid (kWh/m^3_{feed})	5.5	3.5	5.8	15.0	5.5	0
Total fuel import from grid (kWh/m^3_{feed})	133.9	30.4	134.6	13.8	68.2[a]	−19.1[b]
Total electricity self-generation in plant (kWh/m^3_{feed})	0	12.15	0	5.51	0	18.2
Total electricity consumption in plant (kWh/m^3_{feed})	5.51	15.65	5.8	20.5	5.5	18.2
Cooling tower power demand (kWh/m^3_{feed})	2.47	0.40	2.5	0.3	2.6	0.44
Heat pump power demand (kWh/m^3_{feed})	0	13.87	0	18.6	0	16.5
Refrigeration power demand (kWh/m^3_{feed})	3.0	1.38	3.3	1.50	2.9	1.31
Product yield (kg_{prod}/m^3_{feed})	402.1	402.1	359.0	359.0	212.5	216.1

[a] Fuel import can be reduced thanks to the production of biogas from whey biodigestion.
[b] Negative value indicates net export of fuel in the dairy plant scenario with biogas production.

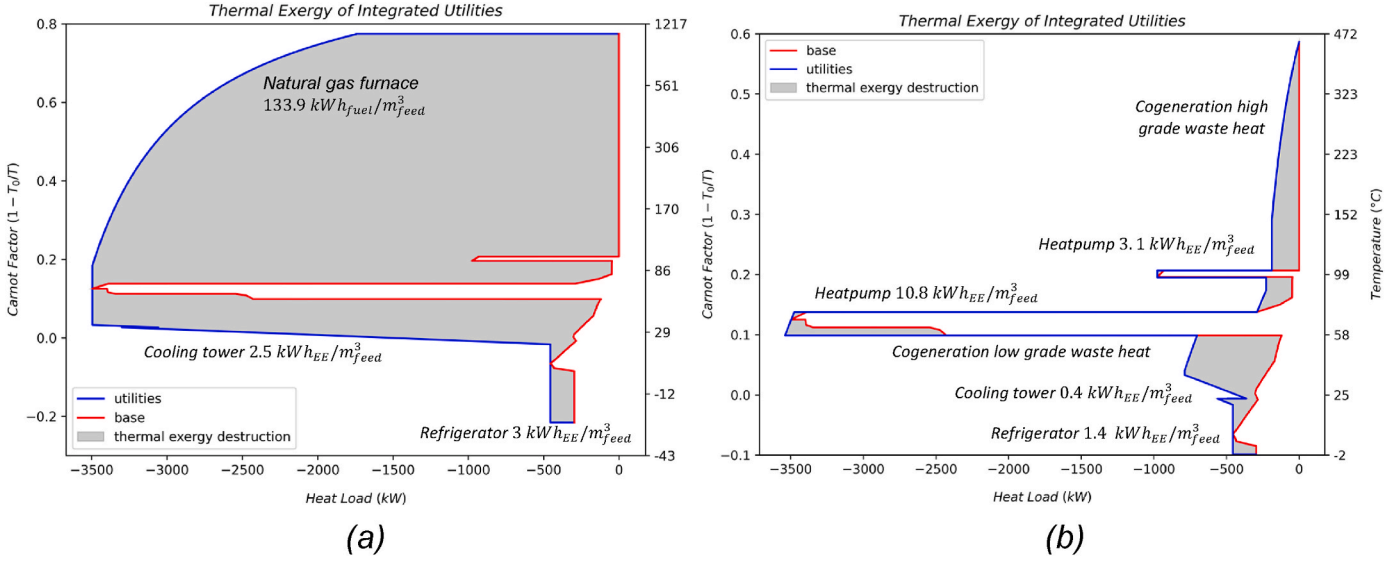


Fig. 10. Dairy plant integrated Carnot plots (ICP) for the mesost production scenario (a) using conventional utilities, (b) using optimized utilities (cogeneration unit and heatpumps).

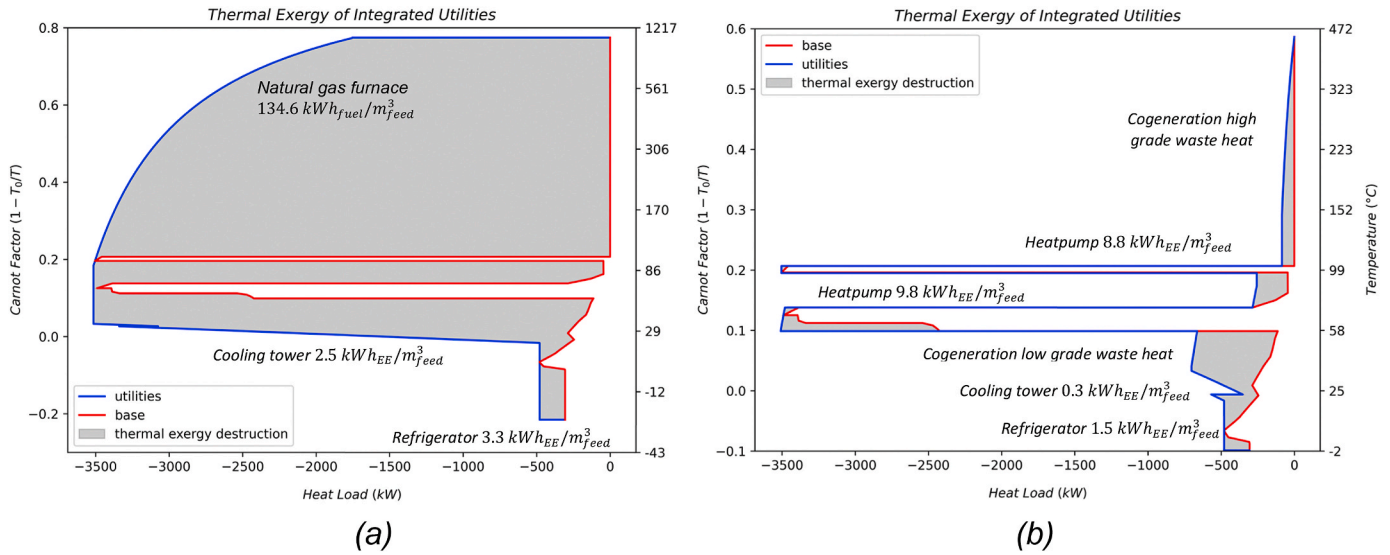


Fig. 11. Dairy plant ICPs for the soft drink production scenario (a) using conventional utilities, (b) using optimized utilities (cogeneration unit and heatpumps).

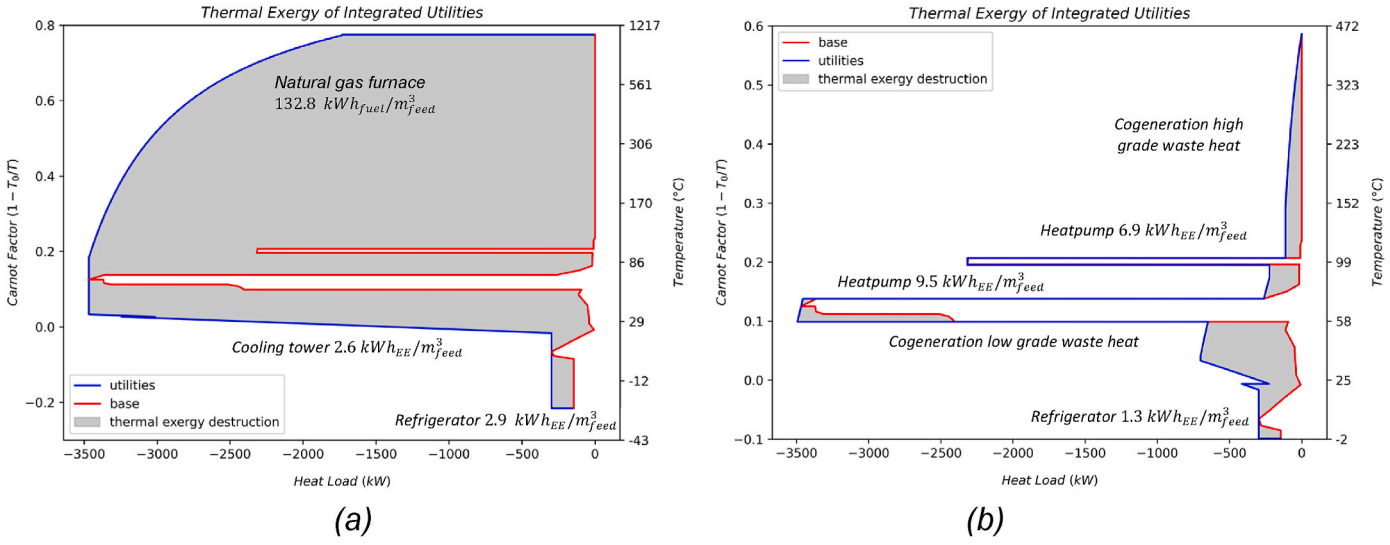


Fig. 12. Dairy plant ICPs for the biogas production scenario (a) using conventional utilities, (b) using optimized utilities (cogeneration unit and heatpumps).

biogas production case. In the latter scenario, the produced biogas is sufficient to supply the onsite electricity demands through a cogeneration unit, and surplus biogas is exported.

3.2. Economic and environmental analyses

Table 4 summarizes the economic and environmental performance indicators, including income rate, operating and capital expenses, net

Table 4
Economic and environmental performance indicators of the different dairy process and whey upgrading routes simulated in ROSMOSE tool.

Process layout	Mesost production Dairy (base case)		Soft drink production (SDP)		Biogas production (BioP)	
	Conventional	Heat pump & Cogeneration	Conventional	Heat pump & Cogeneration	Conventional	Heat pump & Cogeneration
Income (€/m^3_{feed})	6,108	6,108	1,334	1,334	863	863
Operating cost (€/m^3_{feed})	544	529	587	572	544	529
Capital cost[a] (€/m^3_{feed})	0.38	0.49	0.38	0.65	0.38	0.57
Net revenue (€/m^3_{feed})	5564	5579	747	762	319	334
Direct fossil CO_2 emission ($kgCO_2$/m^3_{feed})	26.0	5.9	26.1	2.7	12.9	0
Indirect fossil CO_2 emission ($kgCO_2$/m^3_{feed})	3.23	0.87	3.27	1.23	1.8	0
Net CO_2 emitted ($kgCO_2/m^3_{feed}$)	29.2	6.8	29.4	3.9	14.7	0

[a] Capital cost is annualized assuming a 40-year investment lifetime and 6 % interest rate, and then normalized by the processing rate of milk (28 m^3/h).

revenue, and CO_2 emissions of the various production routes with different utility integration options (e.g., heat pumps and cogeneration units). Carbon taxation was applied to all scenarios and included in the operating expenses of the dairy site. According to these results, due to the high market value of mesost (i.e., brown cheese) and its higher yield compared to the soft drink or biogas production routes, the valorization of the whey stream through mesost production proves to be the most profitable scenario. It is worth mentioning that although the incremental capital cost of installing heat pumps and a cogeneration unit is considered, the net revenues for all the advanced utilities scenarios with mesost, soft drink, and biogas production are slightly higher than when using conventional utilities. As expected, the main contributor to the operating costs (95–97 %) after optimizing the energy integration for all advanced scenarios is the cost of fresh milk imported into the system. Hitherto, a break-even analysis based solely on the price of natural gas would not be sensitive enough to evaluate the effect of fluctuating fossil fuel prices for these advanced decarbonization scenarios. It must be borne in mind that the biogas produced in the whey biodigestion route may fully supersede fossil fuel consumption and even enable exporting the excess of biomethane. Accordingly, additional revenues are envisaged, depending on the whey conversion route studied.

Therefore, a sensitivity study on the overall energy bill (excluding the cost of fresh milk) can be performed as a function of the variable natural gas price. In this sensitivity analysis, two scenarios involving a cogeneration system are evaluated, namely, the mesost and the biogas production options using more advanced utility systems. These two particular scenarios entail a trade-off between using the whey by-product upgrading to produce either goods or energy, as the former one still burns fossil natural gas in a cogeneration unit to reduce the grid electricity import. According to.

Fig. 13, by increasing the price of natural gas from 0.1 to 0.6 €/kWh, the energy bill almost doubled. On the other hand, in the biogas-producing scenario, a natural gas price equal to or higher than 0.6 €/kWh sets a break-even point for which the energy bill becomes null, and the net export of biomethane becomes more and more competitive vis-à-vis market natural gas prices.

Both direct and indirect CO_2 emissions from the integrated dairy plants have been estimated using the ROSMOSE tool. According to Table 4, fossil CO_2 footprint is sharply reduced (90 %) if heat pumps are introduced and electricity is imported from the market or produced in an on-site cogeneration unit. Shifting from the conventional natural gas heater to electricity-driven heat pumps requires much less energy and, therefore, allows optimizing the utilization of waste heat from the process streams. Since the on-site cogeneration unit is powered with biogas from the whey biodigestion, the direct emissions are biogenic, and the dairy plant can become self-sufficient and carbon neutral. These results confirm the promising potential for decarbonizing food industries, such as dairy processing sites, by integrating optimal utility systems.

Fig. 14 (a–b) shows the breakdown of the energy consumption by type (fuel or electricity) and the categorization of the CO_2 emissions (direct or indirect). As a result, a significant reduction in the overall system energy consumption is observed, even for scenarios that reach net negative emissions. Particularly, the biogas production scenario with heat pumps and a cogeneration system is revealed as the most attractive carbon-neutral solution. Yet, this scenario has a drawback of reduced added value (i.e., fuel production) and economic profitability, as a large amount of whey is transformed into biogas instead of mesost.

3.3. *Effect of the full electrification of the heat supply on the performance indicators when using solar electricity generation*

Finally, complete electrification of the most economically profitable scenario (i.e., mesost production) is considered, and the corresponding integrated Carnot plot is shown in Fig. 15. In this alternative mesost production route, heat pumps can fully satisfy the energy needs of the dairy site at the expense of a total electricity consumption of 19.7 kWh_{EE}/m^3_{feed}. If the electrical power that drives the heat pumps is generated by a renewable electricity mix, the installation of a cogeneration system can be spared, reducing by 34 % the capital investment of the utility systems. The net revenue of this scenario remains similar to that of other mesost production cases (~5,580 €/$m^3_{freshmilk}$). This approach contributes to minimizing the irreversibility throughout the industrial process, as can be evidenced by comparing Fig. 15 and Figs. 9–10. Since higher irreversibility is directly linked to improper use of natural resources, its minimization implies not only the reduction of the environmental impact but also the increase of economic attractiveness of the energy systems, as a higher yield of the main product can be achieved using less material and energy input.

The electricity demand of the heat pumps in a fully electrified scenario may be partly supplied by installing photovoltaic panels (PV). Considering a dairy production facility located in the city of Sierre, Switzerland, an average irradiance potential of 163 W_{EE}/m^2_{PV} can be estimated based on the REHO open-source software. The monthly variation is represented in Fig. 16 [78,79]. Meanwhile, considering a

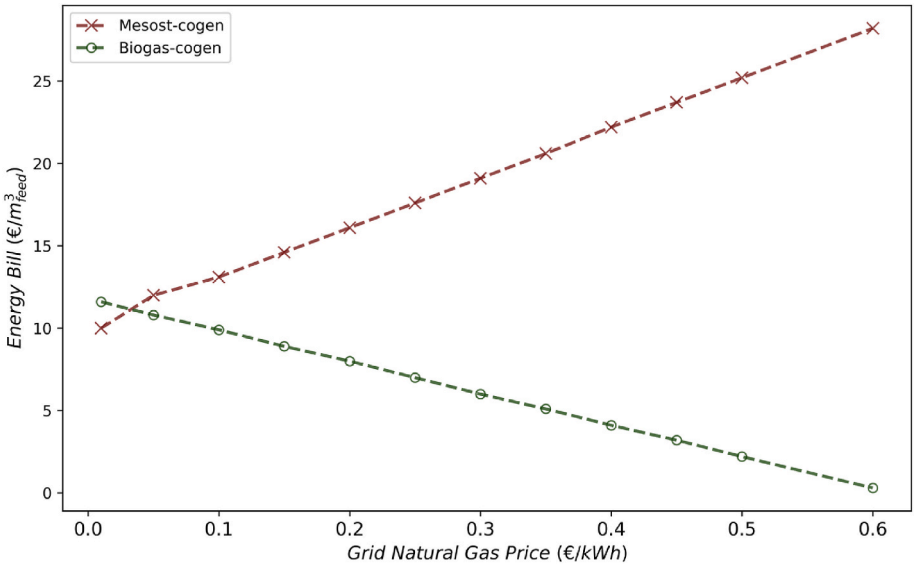


Fig. 13. Energy bill as a function of the market price of fossil natural gas showing the effect of the biomethane production in the biodigestion-based configuration.

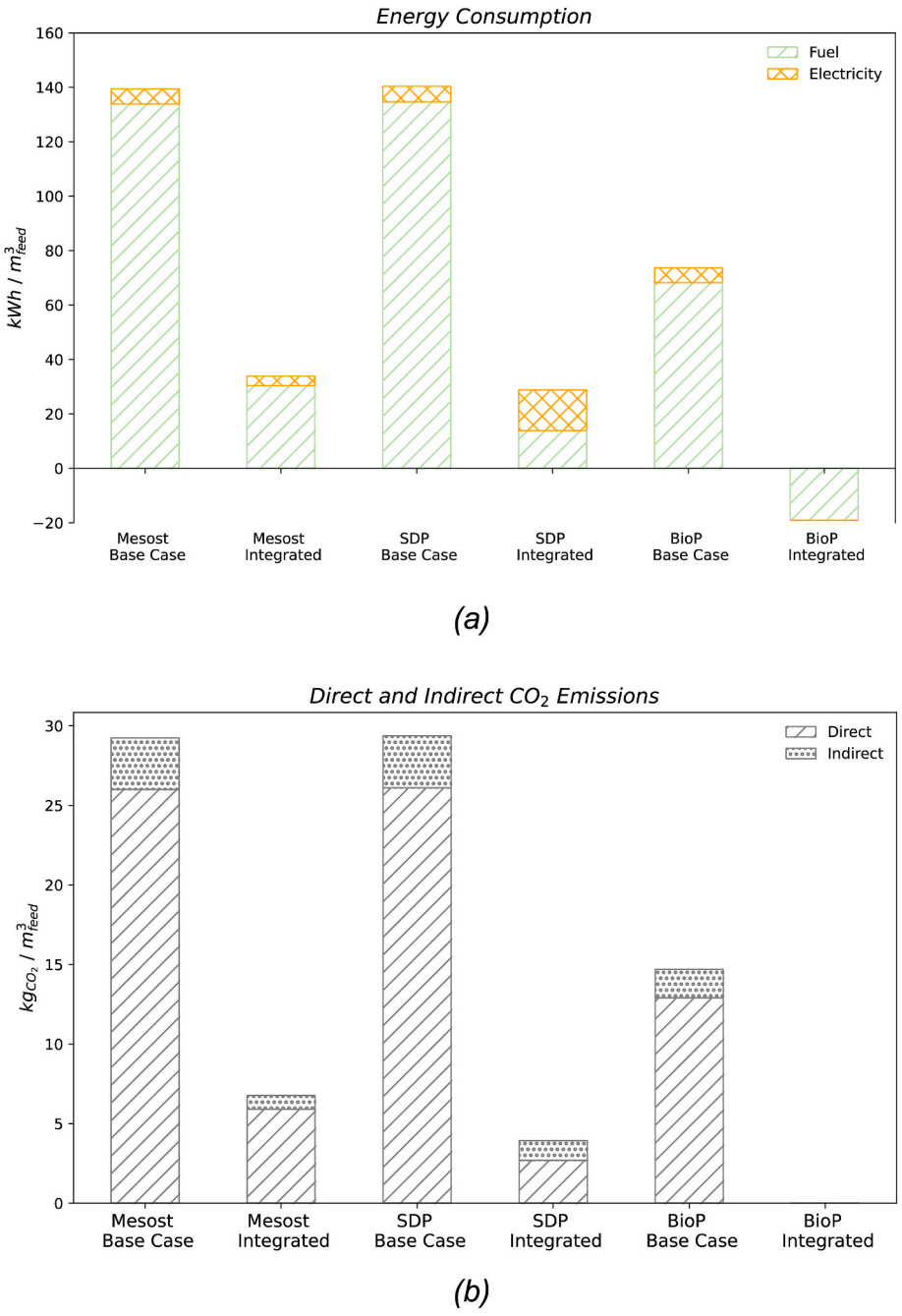


Fig. 14. Summary of (a) net system energy consumption (based on grid import) and (b) environmental impact indicators for the compared process cases.

corrected average PV panel efficiency of ~15–20 % [80] (an area of around 10,140 m^2 of PV panels would be required to supply 60 % of the annual electricity demand for a dairy plant processing 28 m^3/h of milk with whey upgrading process for mesost production, or equivalently, 362 $m^2_{PV}/(m^3_{feed}/hr)$. Fig. 17 shows the breakdown of the seasonal solar power availability, in which it can be evidenced that, during the summer months, solar electricity supplies more than 80 % of the power demand. Meanwhile, during the winter months, significant electricity imports (>80 %) from the grid may be necessary. The investment cost for the PV system is estimated at 0.12 M€/yr, equivalent to 0.50 €/m^3_{feed} [79], similar to other utility systems of the dairy plant, as reported in Table 4.

In summary, the various decarbonization and waste upgrading options for the dairy plant bring about higher energy efficiency through *(i)* enhanced heat integration, *(ii)* advanced utility systems, and *(iii)* renewable energy sources. The ROSMOSE tool and its modeling, optimization, visualization, and reporting functionalities proved critical to assess the different routes and waste management strategies that maximized profitability by producing either mesost cheese or natural gas from upgrading the whey stream. A simple sensitivity analysis of these decarbonization options based on the variation of the fuel energy prices shows a hidden compromise between carbon taxation, biogenic carbon emissions, the fuel energy balance, and the profitability of the dairy site for hypothetical scenarios of sudden natural gas price increments. The implementation of heat pumps for meeting energy demands of the food and beverage sectors at lower temperature levels has a promising potential, especially when coupled with renewable electricity sources, such as PV panels or renewable electricity grids. Similar studies reported an emissions reduction potential of up to 96 % for dairy plants following the installation of heat pumps and the integration of renewable resources [3]. The decarbonization options presented in this work

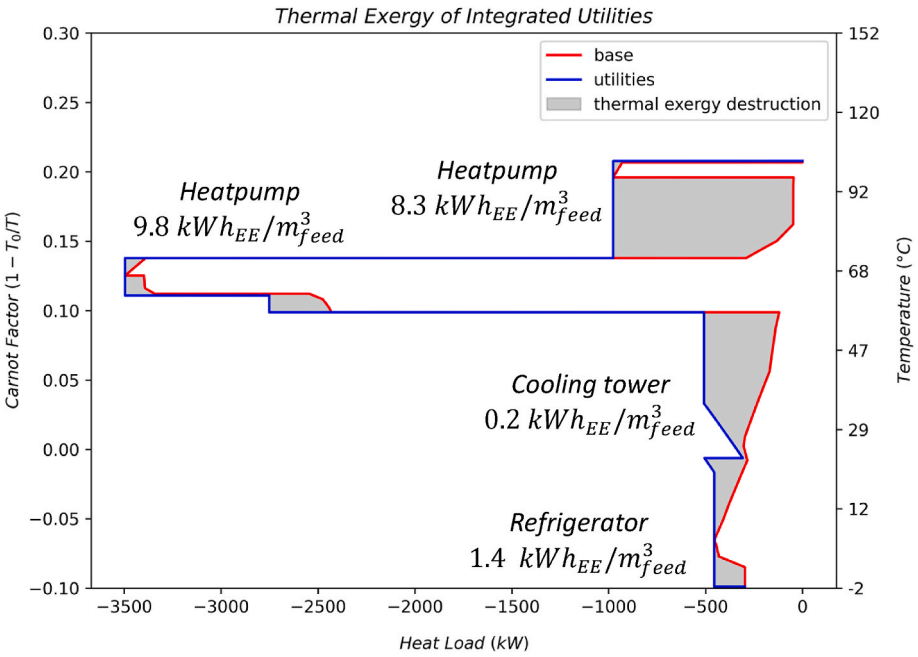


Fig. 15. Integrated Carnot plot of the mesost production route with a full electrification approach.

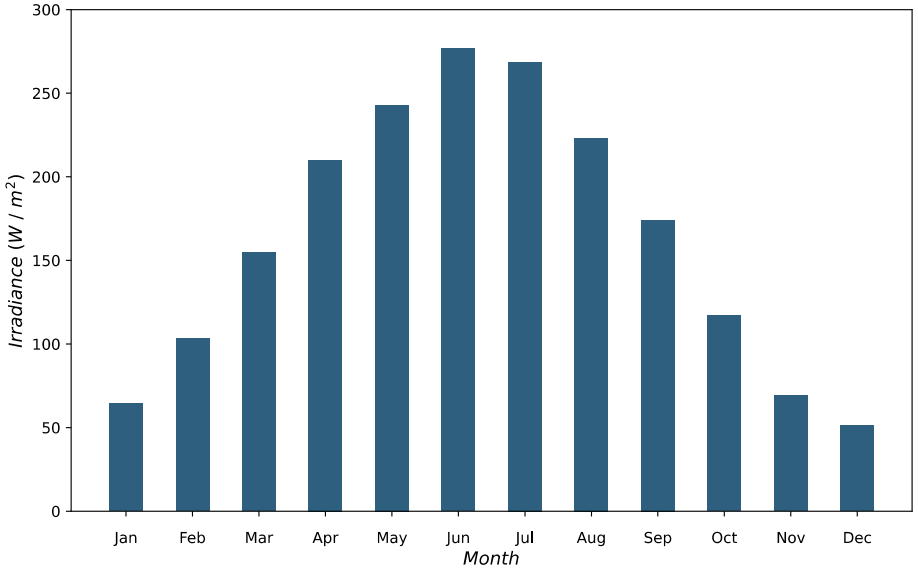


Fig. 16. Monthly solar irradiance in the city of Sierre, Switzerland generated using ROSMOSE tool and REHO platform [81].

using the ROSMOSE tool align with the food industry's strategic roadmaps. In effect, solutions such as switching to biogas obtained from anaerobic digestion and utilization of heat pumps are presented in recent reports [3,82,83] as fundamental efforts towards emissions savings for the dairy industry [83]. However, different from the previous approaches, using the ROSMOSE tool developed in the Quarto environment can help share models, methods, and results more transparently, so that they can be reproducible and verifiable by independent researchers.

4. Conclusions

In this work, a novel process modeling, integration, optimization, visualization, reporting, and documentation tool called ROSMOSE is developed and introduced for its use by a wider scientific community and interested industrial partners. As a demonstrative example, ROSMOSE is applied to design and evaluate decarbonization options for a typical dairy process with waste upgrading pathways. The main outcomes of the study are.

1) The establishment and dissemination of the ROSMOSE platform introduces many features absent in other existing process integration tools. ROSMOSE improves the accessibility of the OSMOSE engine via dynamic reporting environments, such as Quarto, thus enabling the use of a high-level modeling language.
2) Using ROSMOSE simplifies the optimization problem statement and the solution while maintaining the flexibility needed for building energy technology models for entire industrial sites. This capability is often not facilitated in the rigid pre-defined models available in most current proprietary optimization tools.
3) In ROSMOSE the concept of conventional utility units is expanded to superstructures solved as part of the MILP optimization problem.

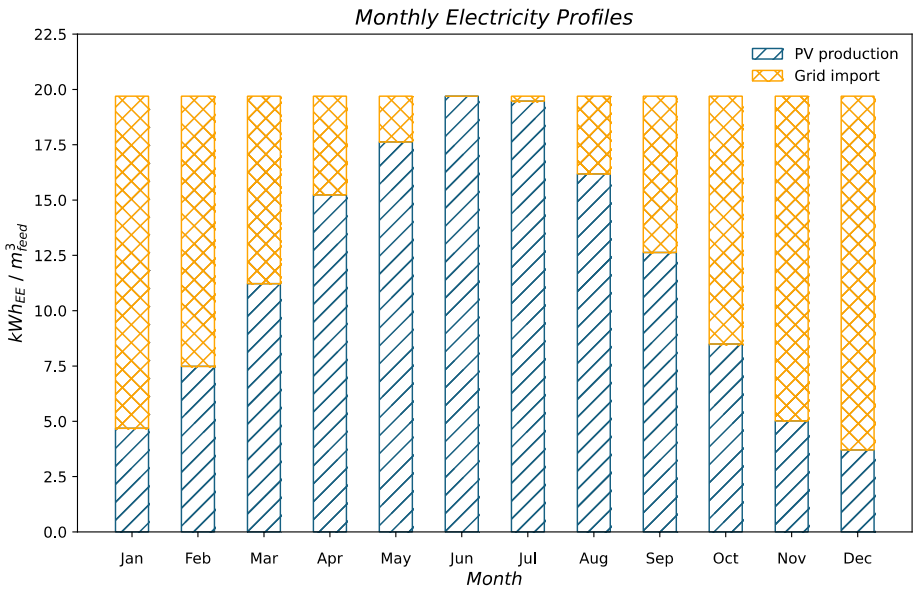


Fig. 17. Breakdown of the monthly electricity demand by type (generation and import) for fully electrified utility system of a dairy plant integrated to a mesost production system.

Those features guarantee the most favorable operating conditions when integrating utilities, such as heat pumps and steam networks, into industrial sites.

4) The ROSMOSE tool modeling functionalities can handle industrial sites and urban systems consistently to any degree of detail specified by the user. Compatible decarbonization solutions for industry and society can be modeled and easily shared among researchers, making scientific reports more reproducible and verifiable.
5) The comparison of the decarbonization routes of the dairy site using the ROSMOSE tool showed that integrating more efficient energy technologies and cleaner energy vectors could drop fuel consumption by 75 %–90 %, compared to conventional utility systems. Also, emissions reduction of up to 96 % is achievable when integrating advanced utilities and waste upgrading options, such as biodigestion, heat pumps, and cogeneration systems.
6) ROSMOSE can be adopted to aid decision-making for the design and optimization of any industrial or urban energy system. Future advancements of the platform include adding more connections to other optimization tools. For instance, EnergyScope [84] allows evaluating energy solutions, on a regional and national level, by taking into consideration multi-energy and multi-sector modeling approaches. Also, REHO tool (Renewable Energy Hub Optimizer) is a decision support tool specialized in urban energy systems planning [81] compatible with the methodology adopted by the ROSMOSE tool.
7) Apart from its multiple applications in research projects [61,62,85–87], the ROSMOSE tool presented in this work is used for imparting different courses at EPFL, which further proves its suitability at both industrial and academic levels. In effect, sharing the ROSMOSE open-source code and the derived reports with the industrial partners and research collaborators only requires cloning the respective GitHub repository, and following the detailed step-by-step installation and use instructions elaborated in the supplementary material SI of this paper.

CRediT authorship contribution statement

Dareen Dardor: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Flórez-Orrego:** Writing – review & editing, Validation, Software, Methodology, Formal analysis, Conceptualization. **Cédric Terrier:** Writing – review & editing, Validation, Software, Methodology. **Meire Ellen Ribeiro Domingos:** Writing – review & editing, Validation, Software, Methodology. **Cyrille Platteau:** Software. **Joao Carlos da Silva:** Software. **Michel Lopez:** Software, Methodology. **François Maréchal:** Writing – review & editing, Supervision, Software, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2024.132182.

Nomenclature

BioP	Biogas production
CC	Composite curve
CAPEX	Capital expenditure
ET	Energy technology
GCC	Grand composite curve
HEN	Heat exchanger network
HLD	Heat load distribution
ICP	Integrated Carnot plot
MER	Minimum energy requirements
MILP	Mixed integer linear programming
OPEX	Operating expenditure
OSMOSE	Multi-Objective Optimization of Integrated Energy Systems
ROSMOSE	Tool and meta-language for high-level interaction with OSMOSE backend engine
SDP	Soft drink production

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